

Probability

Wednesday, 21 April 2021 8:40 AM

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1 Definition

$$\text{Probability} = \frac{\text{No. of favourable outcomes}}{\text{Total outcomes}}$$

2 Q1 — Two coins tossed

(1) Qs: 2 coins are tossed simultaneously, find the probability of getting -

- (i) Head on both coins.
- (ii) None head on both coins.
- (iii) 1 Head and 1 Tail on both coins.
- (iv) At least 1H on both coins.
- (v) At most 1H on both coins.

Total no. of outcomes (4)

$$= \{HH, HT, TH, TT\}$$

- (i) $P(HH) = 1/4$

(ii) $P(TT) = 1/4$

(iii) $P(HT \text{ or } TH) = 2/4 = 1/2$

(iv) $P(\text{at least one head})$

$$= 1 - P(\text{none})$$

$$= 1 - P(TT)$$

$$= 1 - 1/4$$

$$= 3/4$$

(v) $P(\text{At most one head})$

$$= \frac{N((H, T) (T, H) (T, T))}{4}$$

$$P = 3/4$$

3 Q2 — Single dice

(2) DICE:- 1/2/3/4/5/6 0/2/4/6/8

One dice rolled then what is the probability -

(i) even no. comes in front face $\Rightarrow \frac{3}{6} = 1/2$ Ans

[The next two sub-parts continue on a pasted photo of a whiteboard, not in the student's own hand.]

(ii) No. should be divisible by 2 $\Rightarrow \frac{3}{6} = 1/2$ Ans

(iii) No. should be greater than 2 but lesser than 4.

$$\Rightarrow \frac{1}{6} \text{ Ans}$$

4 Q3 — Two dice, sum greater than 9

(3) DICE:-

Ques: Two dice are thrown simultaneously find the probability of getting -

(i) Sum is greater than 9 on the face of both the dice?

Favourable outcomes

$$= \{(6, 4) (4, 6), (6, 5) (5, 6), (6, 6) (5, 5)\}$$

$$P = \frac{6}{6 \times 6} = \frac{1}{6}$$

Total possible outcomes: 6×6 — direct product of space of states.

5 Q4 — Two dice, sum divisible by 5, 4

(4) DICE:-

Ques: Two dice are thrown simultaneously find the probability of getting -

(ii) Sum is divisible by 5.

(iii) Sum is divisible by 4 but not by 3.

(i,i) Sum (6, 1) Case: $2 \leftrightarrow 12 \rightarrow 1, 3 \leftrightarrow 11 \rightarrow 2, 4 \leftrightarrow 10 \rightarrow 3, 5 \leftrightarrow 9 \rightarrow \dots$ (running count of ways to make each total)

(ii) Sum = 5, 10

$$\begin{aligned} &\longrightarrow \{(5, 5) (4, 6) (6, 4)\} \\ &\{(2, 3) (3, 2) (1, 4) (4, 1)\} \\ \text{Probability} &= \frac{3+4}{6 \times 6} = \frac{7}{36} \end{aligned}$$

(iii) Sum is divisible by 4 but not by 3

Required outcomes = 4, 8

$$\begin{aligned} &\longrightarrow \{(4, 4), (5, 3) (3, 5) (2, 6) (6, 2)\} \\ &\{(1, 3) (3, 1) (2, 2)\} \\ \text{Probability} &= \frac{8}{36} = \frac{2}{9} \end{aligned}$$

6 Cards

Total cards = 52 (13 + 13 + 13 + 13)

POKER CARD SYMBOL

Heart	Diamonds	Club	Spade
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$$\text{Face cards (12): } \left. \begin{array}{l} \text{King} \\ \text{Queen} \\ \text{Jack} \end{array} \right\}$$

Ace, together with King/Queen/Jack, $\rightarrow$ Honour Cards (16)

7 Q5 — Probability of drawing a King

(5) Ques: A card is drawn from a pack of 52 cards, find the probability that the card drawn is King?

Solⁿ:

$$P = \frac{4 \text{ kings}}{52 \text{ cards}} = \frac{1}{13}$$

(ii) Find the probability of getting 2 kings when withdrawing 2 cards

$$P = \frac{\text{fav. outcomes}}{\text{total outcomes}} = \frac{{}^4C_2}{{}^{52}C_2}$$

(${}^4C_2 \leftarrow$ No of ways of selecting 2 kings from 4 kings; ${}^{52}C_2 \leftarrow$ No of ways of selecting 2 cards)

$$P = \frac{\frac{4!}{2!2!}}{\frac{52!}{50!2!}} = \frac{4 \times 3 \times 2}{52 \times 51} = \frac{1}{13 \times 17}$$

$$P = \frac{1}{221}$$

8 Q6 — Probability of King or Queen

(6) A card is drawn from a pack of 52 cards. Find the probability that the card drawn is King or Queen?

$$P = \frac{\text{fav. outcomes}}{\text{total outcomes}} = \frac{8}{52} = \frac{2}{13}$$

or

$$P = \frac{\text{No of ways of getting King or Queens}}{\text{No of ways of selecting 1 card from 52}} = \frac{{}^4C_1 + {}^4C_1}{{}^{52}C_1} \\ = \frac{4 + 4}{52} = \frac{2}{13}$$

9 Q7 — One spade and one diamond

(7) Two cards are drawn from a pack of 52 Cards. Find the probability that one card is from Spade and one from Diamond.

$$\text{fav. outcomes} = 13 \times 13 = 169$$

$$P = \frac{\text{fav. outcomes}}{\text{total outcomes}} = \frac{13 \times 13}{{}^{52}C_2} = \frac{{}^{13}C_1 \times {}^{13}C_1}{{}^{52}C_2}$$

$$P = \frac{13 \times 13 \times 2}{52 \times 51} = \frac{13}{102}$$

10 Q8 — Bag of coloured balls

(8) Ques: A bag contains 3 Red balls, 4 Black Balls and 5 green balls. 3 Balls are drawn randomly. Find the probability of getting -

$$3R + 4B + 5G = 12 \quad (3 \text{ balls are drawn randomly})$$

(i) All the balls are different colours.

$$P = \frac{\text{fav outcomes}}{\text{total outcomes}} = \frac{N(\text{All balls of diff colors})}{N(3 \text{ Balls drawn})}$$

12	11	10
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$$\frac{12 \times 11 \times 10}{3!} = {}^{12}C_3 \\ = \frac{N(1 \text{ Red and 1 Black and 1 Green})}{{}^{12}C_3}$$

[As written, this line reads “1 Red and 1 Blue and 1 Green” — a slip for Black, since the bag has no blue balls; parts (ii)–(vi) below correctly call it Black.]

$$= \frac{{}^3C_1 \times {}^4C_1 \times {}^5C_1}{{}^{12}C_3} = \frac{3 \times 4 \times 5 \times 3 \times 2}{12 \times 11 \times 10} \quad (3 \times 2 = \text{No. of permutation of 3 objects})$$

$$\left(P = \frac{3}{11} \right) \checkmark$$

(ii) $P(\text{All the balls are of same colour})$

$$\begin{aligned}
 &= \frac{N(RRR \text{ or } GGG \text{ or } BBB)}{\text{No of total ways of selecting 3}} = \frac{{}^3C_3 + {}^5C_3 + {}^4C_3}{{}^{12}C_3} \\
 &= \frac{1 + \frac{5 \times 4}{2} + 4}{\frac{12 \times 11 \times 10}{3!}} = \frac{15 \times 3 \times 2}{12 \times 11 \times 10} = \frac{3}{44}
 \end{aligned}$$

(iii) $P(\text{None Balls are of Black colour})$

$$\begin{aligned}
 &= \frac{N((2G \& 1R) \text{ or } (2R \& 1G) \text{ or } 3G \text{ or } 3R)}{\text{Total no. of outcomes.}} \\
 &= \frac{{}^5C_2 \times {}^3C_1 + {}^3C_2 \times {}^5C_1 + {}^5C_3 + {}^3C_3}{{}^{12}C_3} \\
 &= \frac{\frac{5 \times 4}{2} \times 3 + 3 \times 5 + \frac{5 \times 4}{2} + 1}{\frac{12 \times 11 \times 10}{3 \times 2}} = \frac{(30 + 15 + 10 + 1) \times 6}{12 \times 11 \times 10} \\
 &P = \frac{56 \times 6}{12 \times 11 \times 10 \times 5} = \frac{14}{55}
 \end{aligned}$$

(OR) $P(\text{None Balls are Black}) = \frac{{}^8C_3}{{}^{12}C_3} = \frac{14}{55}$

(iv) $P(\text{All the Balls should not be black})$

$$\begin{aligned}
 &= 1 - P(\text{All balls Black}) \\
 &= 1 - \frac{BBB}{{}^{12}C_3} = 1 - \frac{{}^4C_3}{{}^{12}C_3} \\
 &P = 1 - \frac{4 \times 3 \times 2}{12 \times 11 \times 10} = 1 - \frac{1}{55} = \frac{54}{55}
 \end{aligned}$$

(v) $P(\text{At least one ball is green}) = 1 - P(\text{No green balls})$

$$\begin{aligned}
 &= 1 - \frac{{}^7C_3}{{}^{12}C_3} \\
 &P = 1 - \frac{7 \times 6 \times 5 \times 3 \times 2}{3 \times 2 \times 12 \times 11 \times 10} \\
 &P = 1 - \frac{7}{44} = \frac{37}{44}
 \end{aligned}$$

(vi) $P(\text{At most one ball is green})$

$$\begin{aligned}
 &= 1 - P(2 \text{ balls green}) - P(3 \text{ balls green}) \\
 &= 1 - \frac{N((2G \& 1B) + (2G \& 1R))}{{}^{12}C_3} - \frac{{}^5C_3}{{}^{12}C_3} \\
 &= 1 - \frac{{}^5C_2 \times {}^4C_1 + {}^5C_2 \times {}^3C_1 + {}^5C_3}{{}^{12}C_3} \\
 &= 1 - \frac{\frac{5 \times 4}{2} \times 4 + \frac{5 \times 4}{2} \times 3 + \frac{5 \times 4}{2}}{\frac{12 \times 11 \times 10}{6}}
 \end{aligned}$$

$$= 1 - \frac{(40 + 30 + 10) \times 6}{12 \times 11 \times 10} = 1 - \frac{80 \times 6}{12 \times 11 \times 10}$$

$$\left(P = 1 - \frac{4}{11} = \frac{7}{11} \right) \checkmark$$

(OR) $P(\text{At most one ball is green}) = \frac{N((1G \& 2 \text{ other}) + (0G \& 3 \text{ other}))}{\text{Total outcomes}}$

$$= \frac{{}^5C_1 \times {}^7C_2 + {}^5C_0 \times {}^7C_3}{{}^{12}C_3}$$

$$= \frac{5 \times 7 \times 6/2 + 1 \times 7 \times 6 \times 5/6}{{}^{12}C_3}$$

$$\left(P = \frac{140}{220} = \frac{7}{11} \right) \checkmark$$